ANSWERS

Chapter 1 Numbers

Exercise 1.1

1. i) 12 ii) 31 iii) 3 iv) 2 v) 10
2. i) False ii) False
   iii) True iv) True v) True
3. smallest → 11; biggest → 97
4. smallest → 100; biggest → 999
5. True. 3 + 7 + 9 = 19 is odd
6. (17, 71),(37,73) & (79,97)
7. False. 9 is odd number but not prime
8. True. The composite number 4 has 3 factors namely 1, 2, and 4.
9. 6, 12, 18, 24, 30 (excluding February)
10. 19
11. a) 60 = 2 x 2 x 3 x 5
    b) 128 = 2 x 2 x 2 x 2 x 2 x 2 x 2
    c) 144 = 2 x 2 x 2 x 2 x 3 x 3
    d) 198 = 2 x 3 x 3 x 11
    e) 420 = 2 x 2 x 3 x 5 x 7
    f) 999 = 3 x 3 x 3 x 37
12. (11, 13) or (13, 11)

Objective Type Questions

13. b) 2 14. c) 2 15. b) 92 16. c) 40
17. a) 80 18. d) impossible 19. a) 2
20. d) all of these

Exercise 1.2

1. i) 15 ii) 2
   iii) 3 iv) 156 v) 3
2. i) False ii) True
   iii)True iv) False v)True
3. i) 6 ii) 17
   iii) 1 iv) 12
   v) 9 vi) 5
4. i) 18 ii) 24
   iii) 30 iv) 42
   v) 120 vi)75
5. HCF → 22; LCM → 18018
6. HCF=20 litres
7. After 360 seconds (6 min), at 8.06 a.m
8. 2 pairs possible
9. 24

Objective Type Questions

10. c) 71, 81 11. d) 9936
12. b) 36 13. c) 80

Exercise 1.3

1. 4 = 2 + 2; 6 = 3 + 3; 8 = 3 + 5;
   10 = 3 + 7 (or) 5 + 5; 12 = 5 + 7;
   14 = 7 + 7 (or) 3 + 11;
   16 = 5 + 11 (or) 3 + 13
2. Yes, because it has only two factors.
3. For n = 2, 3, 4, 6 and 7
4. a) False, 3 is a factor of 9
   b) True, 12 is a multiple of 6
5. i) 8 ii) 0 iii) 9 iv) 1 v) 8
6. False. 12 is divisible by both 4 and 6 but not by 24
7. True. 17+19=36 is divisible by 4
8. 40 cm

Challenge problems

9. 2, 37, 41
10. 11, 13, 17, 19; The sum 60 is divisible by 1, 2, 3, 4, 5 and 6 but not divisible by 7, 8 and 9
11. 2520
12. Yes. 2 x 3 x 4=24 is divisible by 6.
13. Once in 30 days, 31st October
14. The lifts will stop at floors 15, 30, 45, 60, 75, 90 and 105
15. (15, 20)
16. Yes. Since it is divisible by both 8 and 11 and hence by 88
17. After 60 minutes, at 8 a.m

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